

ENT 201
BUG-IN-CUP EXPERIMENT AND JOURNAL
SPRING 2023

You will be given two small, covered petri dishes each containing two small bugs, some food (unsalted sunflower seeds), and tube of water with a cotton wick. The lid on each dish should be secured with a rubber band. Please follow the instructions below for our “bug-in-a-dish” experiment.

1. **Carefully** bring your bugs home. **Don't let the seeds or water tube roll over and kill the little critters!**
2. Place both dishes in a location where you can easily observe them every day and where they will not be crushed by your room mate, eaten by your dog, thrown out by your mother, etc. It is also important that the dishes get neither too hot or too cold; don't set them where sunlight streaming through a window will hit them directly, or you'll cook your bugs. A good place might be your desk or the area where you study. **Keep the lids secured by the rubber bands at all times except as indicated below.**
3. The water in the tubes will be consumed by the bugs and will also evaporate. Thus, it is necessary to refill the tubes every few days. **Don't let the tubes go dry!**
4. One of the dishes will be your untreated “control” for your experiment. For this treatment, refill the water tube as needed (see below) with plain water for the duration of the experiment.
5. The other dish will represent your “experimental” treatment in this experiment. The experimental variable we are going to manipulate is the liquid the bugs have available to drink. Our hypothesis is that giving these insects some liquid other than water will have an effect of some kind on their growth and development. In this dish, when the water in the tube runs out the first time, you will replenish the tube with the liquid of your choice, and in this dish, you will use the same liquid every time the tube needs to be refilled. Be sure to set aside a small quantity of your experimental liquid, keep it in a capped container, and keep it refrigerated to prevent the growth of mold. **Don't use orange juice- it molds too quickly. If you choose a carbonated beverage, open the container and let it stand a few days (in the refrigerator) to “debubble” it before you use it.** *Begin using your experimental liquid with the first dish where the water runs out. Mark this dish “Experimental” and the other “control” at this time. Remember, the “control” dish only gets water for the duration of the experiment.*

Use this procedure to establish and maintain your experimental and control treatments:

- a. Remove the rubber band and carefully remove the tube so that bugs are not injured or allowed to escape. Should they escape, simply allow them to crawl on to a small piece of paper and then tap them gently back into the dish.
- b. You will notice a small piece of plasticine clay on the side of the tube. This was used to help keep the tube from rolling around in the dish. When you replace the tube after filling, make sure the clay side goes on the bottom of the dish
- c. Pull the wick from the tube with a tweezers or pin. If it is moldy, wash it with water.
- d. Fill the tube with the appropriate liquid and replace the wick so that the top of the wick is flush with the top of the tube. Dry off any excess fluid on the outside of the tube!
- e. Return the tube to the dish, being careful not to crush or injure its inhabitants. Put the rubber band back on the dish.

The web address of a video demonstration of this process will be provided to you when the bugs go out.

6. Make daily (more or less) observations of the CONTROL and EXPERIMENT bugs/cups and record your observations in a journal or diary format. These should be your handwritten “lab notes” which should not be typed or reworked before you hand them in. However, please write legibly! **Pretend you are Aristotle, the first person to ever see this critter. Your notes should allow anyone reading them to get a very good picture of your insect and its growth and behavior.**
7. If your control insects die, see the TA's responsible for this assignment, Tay and Rachel, and they'll give you new control insects. It is unusual for the control insects to die, so if yours do, inspect the area you're keeping them in to be sure it is not too hot or cold! If your experimental insects die, see the TA's for replacements and try the experimental liquid again; however, you may want to dilute your experimental liquid when you retry it. Make sure that you enter these experiences in your journal should they occur.

8. Your experiment is complete when your control insects have reached adulthood and you have waited about two weeks to see if your experimental insects complete development, in those cases when your treatment slowed development. At this time, you should record some conclusions about the experiment based on your observations. You should also identify your insects with the aid of the books we bring to class for that purpose.
9. When the insects have reached adulthood, you should use the books made available before and after class to help you identify your bugs. You will also be given further instructions on how to care for your adults.
10. When everyone's experiment is complete you will be asked to turn in your journal. Your grade (50 points or 5% of your grade) will be based on the thoroughness of your observations, the correct identification of your critters, and the conclusions you draw from your observations.
11. If you have any questions during the experiment, contact the grad TA's!
12. Finally, you should read the article that follows about Francesco Redi's experiments on spontaneous generation of insects as a good example of the scientific method.